



NAME – SUBASH G **COURSE NAME – Operation System**
REG NO – 192572150 **COURSE CODE – CSA0408**

Lab Observations

EXPERIMENT - 4

Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next

PROGRAM:

```
#include <stdio.h>

int main()
{
    int A[100][4];
    int i, j, n, total = 0, index, temp;
    float avg_wt, avg_tat;

    printf("Enter number of process: ");
    scanf("%d", &n);

    printf("Enter Burst Time:\n");

    for(i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &A[i][1]);
        A[i][0] = i + 1;
    }

    for(i = 0; i < n; i++)
    {
        index = i;
```

```
    for(j = i + 1; j < n; j++)
    {
        if(A[j][1] < A[index][1])
            index = j;
    }

    temp = A[i][1];
    A[i][1] = A[index][1];
    A[index][1] = temp;

    temp = A[i][0];
    A[i][0] = A[index][0];
    A[index][0] = temp;
}

A[0][2] = 0;

for(i = 1; i < n; i++)
{
    A[i][2] = 0;

    for(j = 0; j < i; j++)
        A[i][2] += A[j][1];

    total += A[i][2];
}

avg_wt = (float)total / n;

total = 0;

printf("\nP BT WT TAT\n");

for(i = 0; i < n; i++)
{
    A[i][3] = A[i][1] + A[i][2];

    total += A[i][3];

    printf("P%d %d %d %d\n",
           A[i][0], A[i][1], A[i][2], A[i][3]);
}
```

```
    avg_tat = (float)total / n;

    printf("\nAverage Waiting Time = %f", avg_wt);
    printf("\nAverage Turnaround Time = %f", avg_tat);

    return 0;
}
```

OUTPUT:

